

CO-ATTENDANCE TO COMMUNITY:

Simulating Analog Social-Network Formation
in Boutique Fitness

Alexander Garcia

DSAN 6400

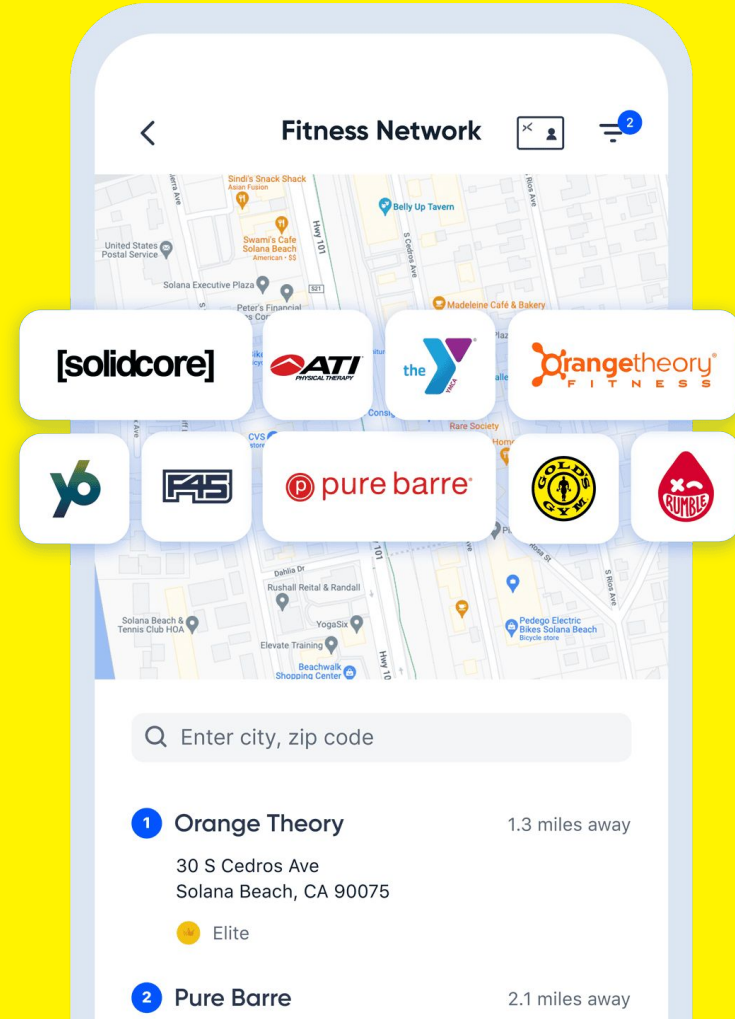
6 August 2026



KEY RESEARCH QUESTION

How does repeated analog co-attendance within SoulCycle form rider networks in the absence of a digital social platform?

- In the digital age, SoulCycle has shied away from a reliance on digitally integrated infrastructures.
- Rather than connecting riders through leaderboards, digital profiles, or an explicit online social network, they center on repeated in-person attendance at particular studios, with preferred instructors and class times.
- This project represents that environment as a temporal, weighted network and examines network structure, rider connectivity, tie strength, and formation mechanisms.



WHAT DOES PRIOR RESEARCH SUGGEST?

The R/Evolution of Civic Engagement (Yuan, Miao, and Liu 2019) - The researchers analyze the Facebook groups frequented by some random 600 members of Occupy Chicago. - Individuals initially connected via one shared interest may also participate in broader communities that extend beyond the original source of affiliation. - However, the larger network remained loosely connected, with a density of 0.02, and highly fragmented, with a modularity score of 0.84.	Adding Linking Value to the Value Proposition (Bàres and Nemani 2023) - Under the linking value framework, customers may patronize a business not only because they value the product being sold, but also because they value the community and shared experience surrounding it. Bàres and Nemani identify SoulCycle as a particularly strong example of this model. - The studio environment and instructional philosophy transform the class from a simple 45-minute cycling workout into a collective experience built around synchronicity, instructor motivation, and shared intensity.
Assessing Social Network Influences on Adult Physical Activity Using Social Network Analysis (Prochnow and Patterson 2022) - Examination of 28 studies of adult physical activity, with the review suggesting that network size, composition, connectivity, and exposure to active peers may shape an individual's likelihood of remaining physically active. - However, much of the existing literature examines how a small personal network relates to one individual's behavior rather than how physical activity operates across a complete network.	Sick of Swiping, Some Daters Would Rather Sweat Together (Friedman 2026) - Friedman provides several examples of fitness brands collaborating with wellness-based dating apps facilitate customer-to-customer interaction. - As Gen Z spends more time and money on fitness and less on alcohol and nightlife, gyms and boutique studios may increasingly function as alternative social gathering spaces and dating locales.

Research Gap: Literature suggests that fitness participation can create **linking value** and that network structure matters for physical activity, but less is known about how **analog** attendance patterns generate those networks over time.

FROM CO-ATTENDANCE TO SOCIAL CONNECTION

Shared class: → **Repeated co-attendance:** → **Familiarity tie:** → **Social tie:** → **Coordinated attendance:**

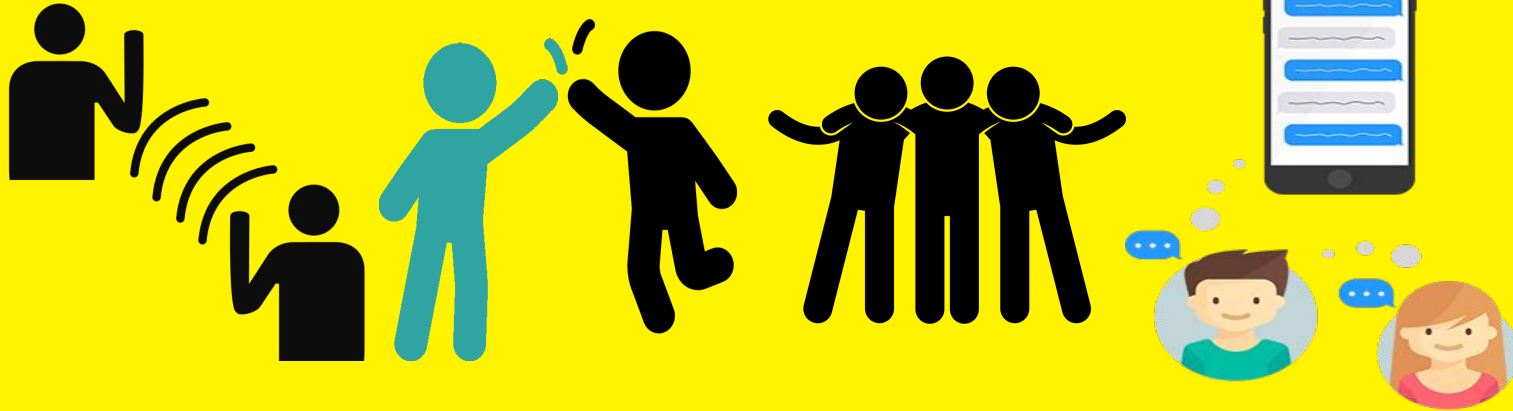
Two riders attend the same class.

The pair encounters one another across multiple classes.

Repetition creates recognition.

Continued interaction forms a categorically stronger connection.

Established social ties influencing future attendance.



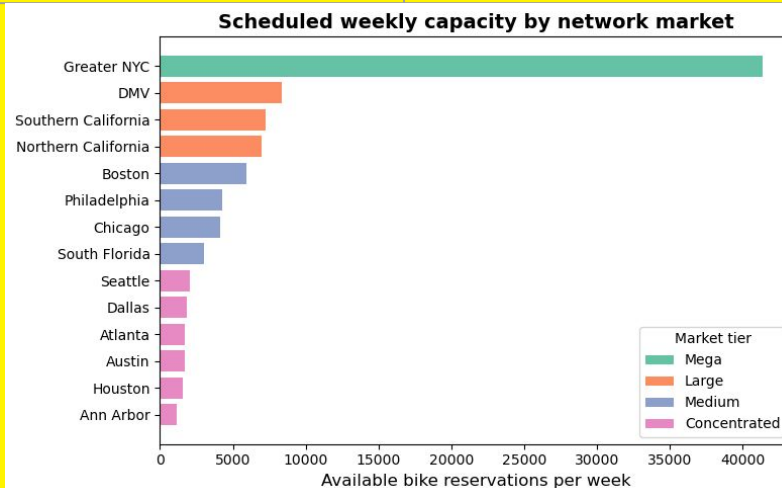
DATA AND SIMULATION ENVIRONMENT

Model Inputs

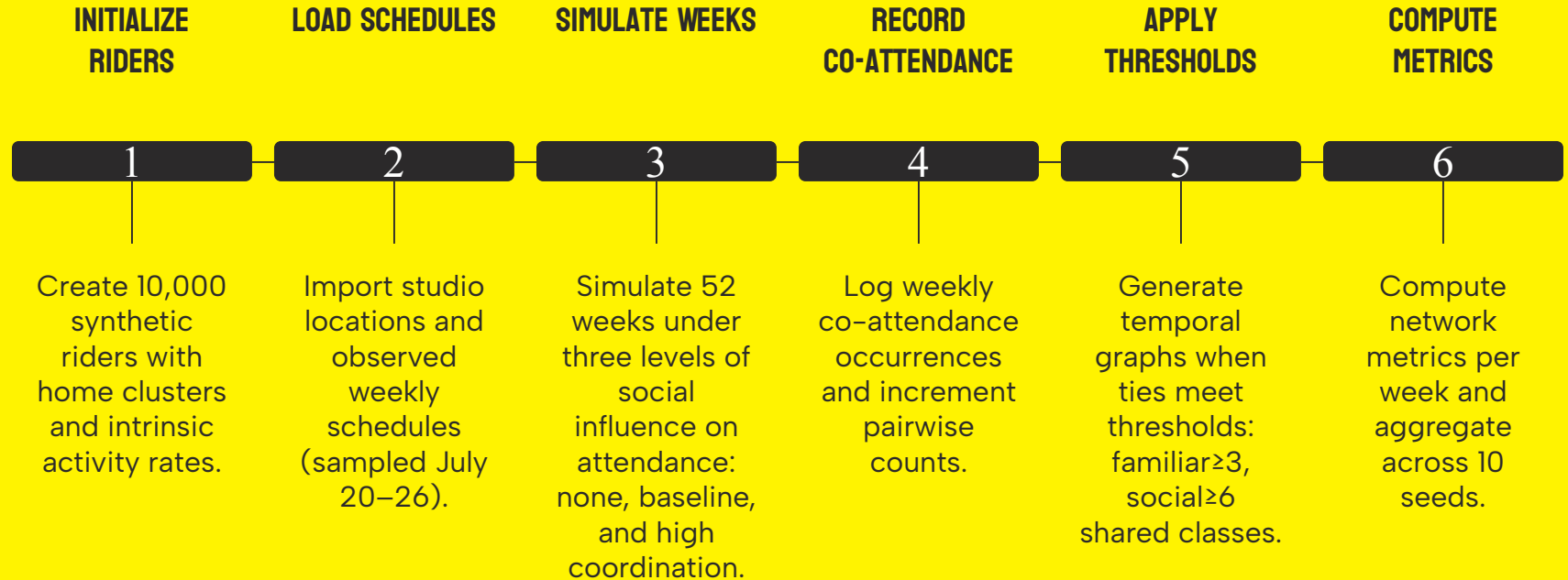
- Publicly available studio locations
- Instructor assignments and class schedules
- Reservable-bike capacity
- Research-defined markets and geographic clusters
- Rider preferences for studios, instructors, days, and times

Simulation Design

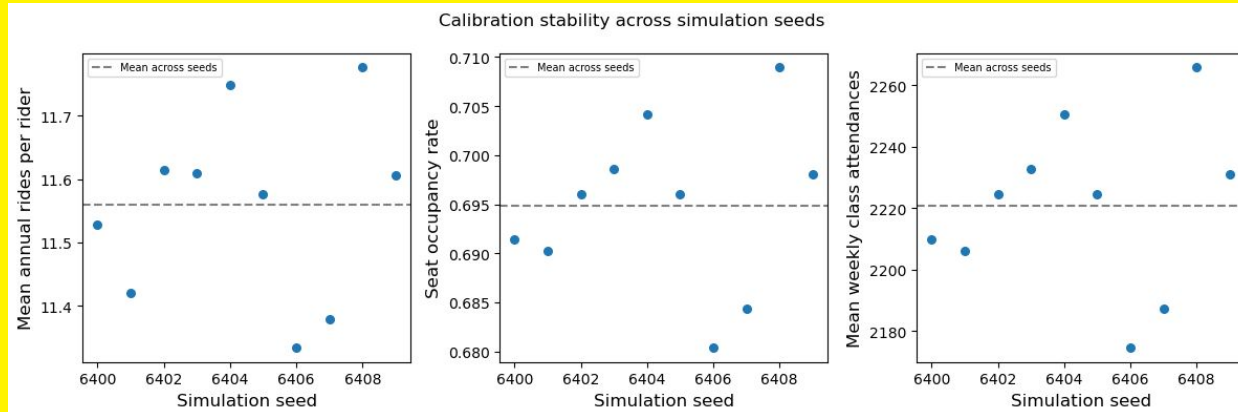
- 10,000 synthetic riders
- 52 simulated weeks
- 10 random seeds
- 3 coordination scenarios (baseline, no coordination, high coordination)
- One class per rider per day



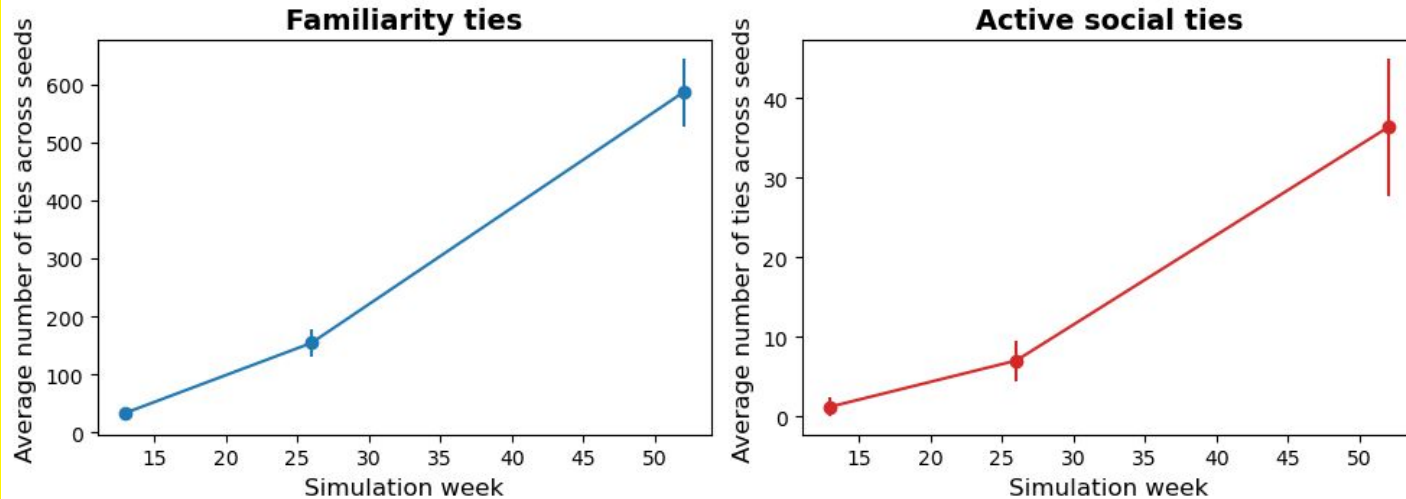
SIMULATION PIPELINE



MODEL VALIDATION AND NETWORK GROWTH

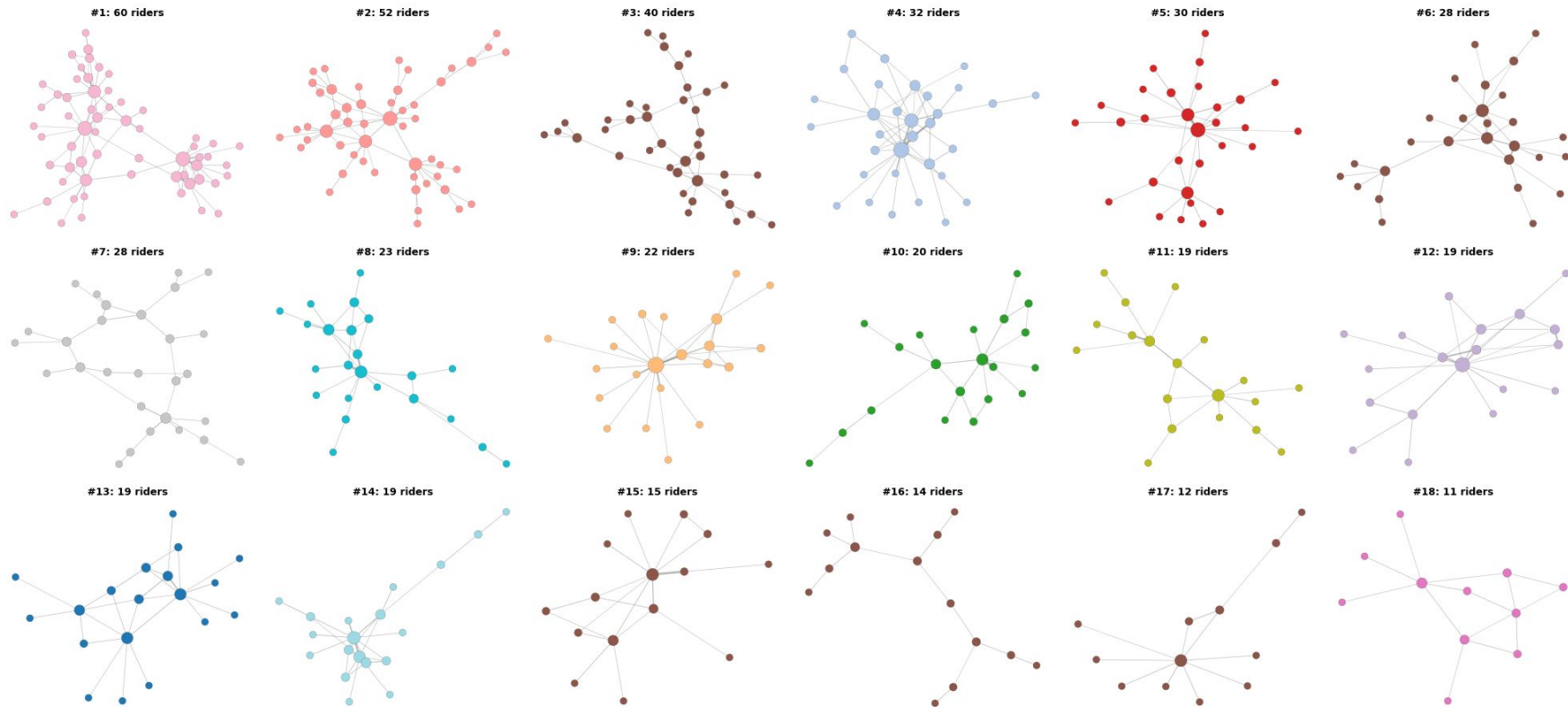


Familiarity grows steadily, while active social ties remain rare



NETWORK FORMATION

Largest local familiarity communities at week 52

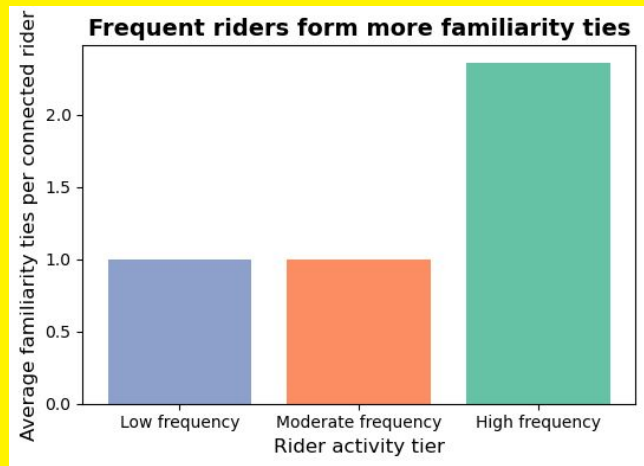
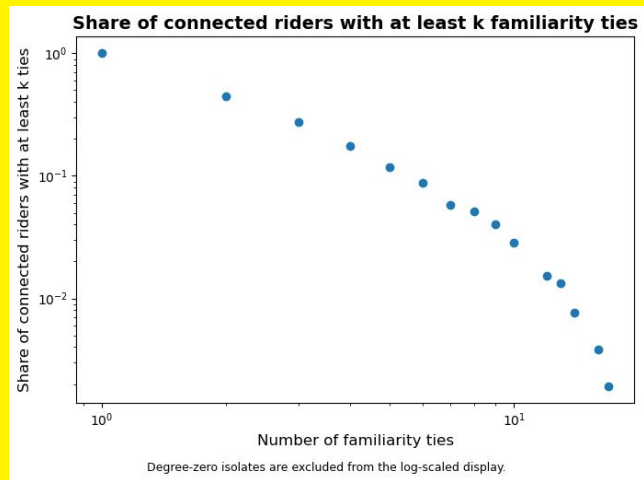


Baseline seed 6400; node size represents degree and edge width represents tie strength. Panels show the 18 largest of 43 components among riders with at least one familiarity tie.



NETWORK METRICS

Network metric	No coordination	Baseline	High coordination
Nontrivial components	45.4 ± 5.9	49.6 ± 9.1	48.3 ± 6.3
Largest component	65.2 ± 15.6	55.0 ± 17.0	54.4 ± 11.0
Largest-component share	0.65 ± 0.16%	0.55 ± 0.17%	0.54 ± 0.11%
Density	0.00128%	0.00117%	0.00117%
Clustering coefficient	0.69 ± 0.12%	0.54 ± 0.09%	0.50 ± 0.10%
Degree assortativity	-0.172 ± 0.068	-0.164 ± 0.063	-0.162 ± 0.055
Modularity	0.932 ± 0.006	0.936 ± 0.007	0.936 ± 0.006

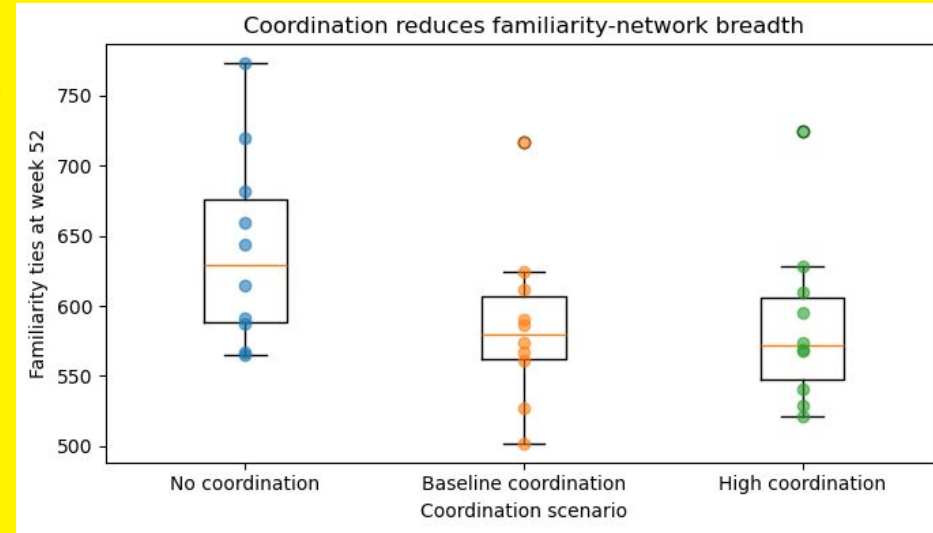


THE COORDINATION DILEMMA

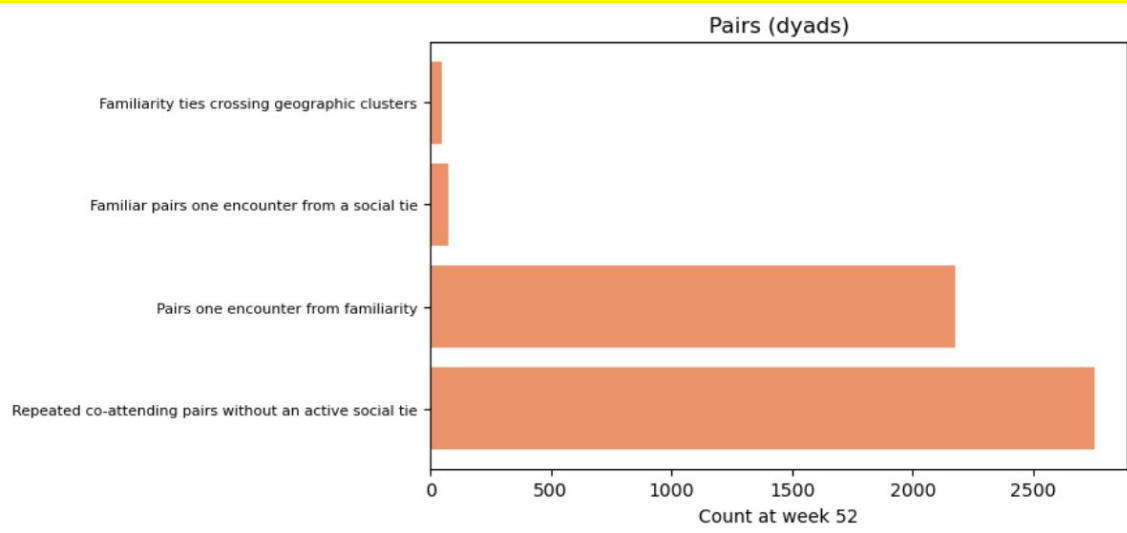
The three coordination rules:

1. **No coordination:** Riders choose classes independently based only on their existing market, studio, instructor, day, and time preferences.
2. **Baseline:** Riders with established active social ties may coordinate with **up to two partners** and attend **up to two coordinated classes per week**. Their remaining attendance choices are made independently.
3. **High coordination:** Riders may coordinate with **up to four partners** and attend **up to four coordinated classes per week**. This represents a stronger feedback loop from the social network back into attendance behavior.

Outcome: Stronger coordination concentrated riders within existing social routines, reducing exposure to new relationships.



LIMITATIONS AND DIGITAL OPPORTUNITY

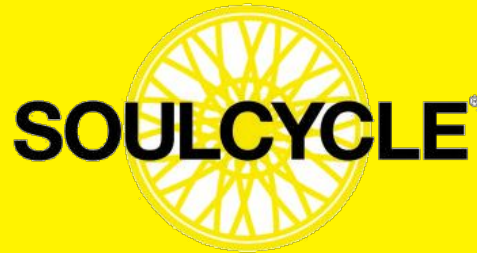


- Thousands of repeated co-attendance relationships form, but few become active social ties. Most of the network's unrealized potential lies between repeated recognition and durable social connection.
- A digital feature would be most useful if it helped riders recognize and coordinate with people they already encounter.

- **Data construction:** The model relies on public schedules, studio locations, instructor assignments, and reservable-bike capacity rather than actual booking histories.
- **Temporal limitation:** One observed schedule week is used as the basis for a 52-week simulation.
- **Behavioral outcomes:** The simulation measures network formation, not whether stronger ties increase attendance, loyalty, or long-term physical activity.
- **Analog vs. digital:** There is no explicit digital mechanism for riders to identify one another, exchange contact information, join groups, etc.
 - The results therefore show the latent opportunity for digital support, not the effectiveness of any particular product.

CONCLUSIONS

1. When simulating, analog co-attendance does generate meaningful network structure.
2. Existing networks likely remain sparse, fragmented, and strongly localized by market.
3. High-frequency riders become more connected, but widespread durable social ties remain rare.
4. Digital integration could strengthen existing offline familiarity rather than replace SoulCycle's analog model with a conventional social network.



REFERENCES

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